MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, FIELD EFFECT TRANSISTOR, N-CHANNEL, SILICON
TYPES 2N6660 AND 2N6661
JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a N-channel, enhancement mode, low-threshold logic level, high frequency, high switching speed MOSFET, power transistor intended for use in high density power switching applications. Four levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1, TO-205AD.

1.3 Maximum ratings.

Type	P_T 1/ T_C = +25°C	P_T T_A = +25°C	V_{DS}	V_{DGR} 3/	V_{GS}	I_{D1} 2/ T_C = +25°C	I_{D2} T_C = +100°C	I_S 2/	I_{DM}	T_{STG}	T_{op}
	W	mW	V dc	V dc	V dc	A dc	A dc	A dc	A(pk)	°C	°C
2N6660	6.25	725	60	60	±20	.99	0.62	-.99	3	-65 to +175	-65 to +150
2N6661	6.25	725	90	90	±20	.86	0.54	-.86	3		

1/ Derate linearly 0.05 W/°C for T_C > +25°C.

2/ Derate above T_C = +25°C according to the formula $I_D = \sqrt{\frac{P(rated)}{K}}$, where P(rated) = 150 -(T_C -25)(0.05) watts; K = maximum $r_{DS(on)}$ at T_J = +150°C.

3/ $R_{GS} \leq 1$ MΩ.

1.4 Primary electrical characteristics at T_C = +25°C.

Type	Min $V_{(BR)DSS}$ V_{GS} = 0 V I_D = 10 µA dc	$V_{GS(th)1}$ $V_{DS} > V_{GS}$ I_D = 1.0 mA		Max I_{DSS1} V_{GS} = 0	Max $r_{DS(on)}$ 1/ V_{GS} = 10 V dc		$R_{\theta JC}$ max
				V_{DS} = 80% of rated V_{DS}	T_J = +25°C at I_{D1}	T_J = +150°C at I_{D2}	
	V dc	V dc Min	V dc Max	µA dc	Ohms	Ohms	°C/W
2N6660	60	0.8	2.0	1	3.0	6.33	20
2N6661	90	0.8	2.0	1	4.0	8.44	20

1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center, RBE-2, Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of the specification and standard required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and figure 1 herein.

3.3.1 Lead material and finish. Lead material shall be Kovar or Alloy 52, a copper core is permitted. Lead finish shall be gold, tin, or solder. Where a choice of lead material or finish is desired, it shall be specified in the contract or purchase order (see 6.3).

3.3.2 Internal construction. Multiple chip construction shall not be permitted.

3.4 Marking. Marking shall be in accordance with MIL-S-19500. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the transistor.

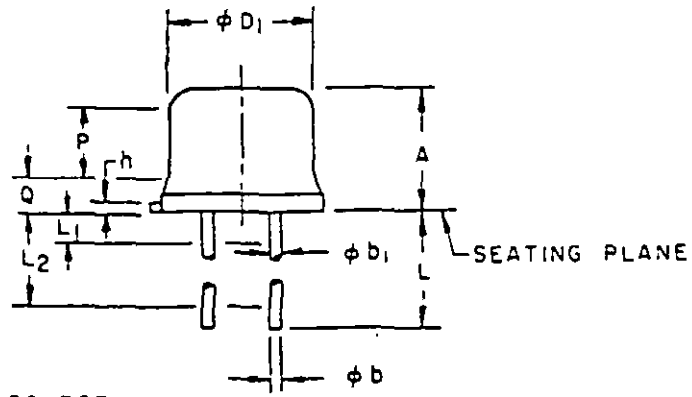
3.5 Electrostatic discharge protection. The devices covered by this specification require electrostatic protection (see 6.2).

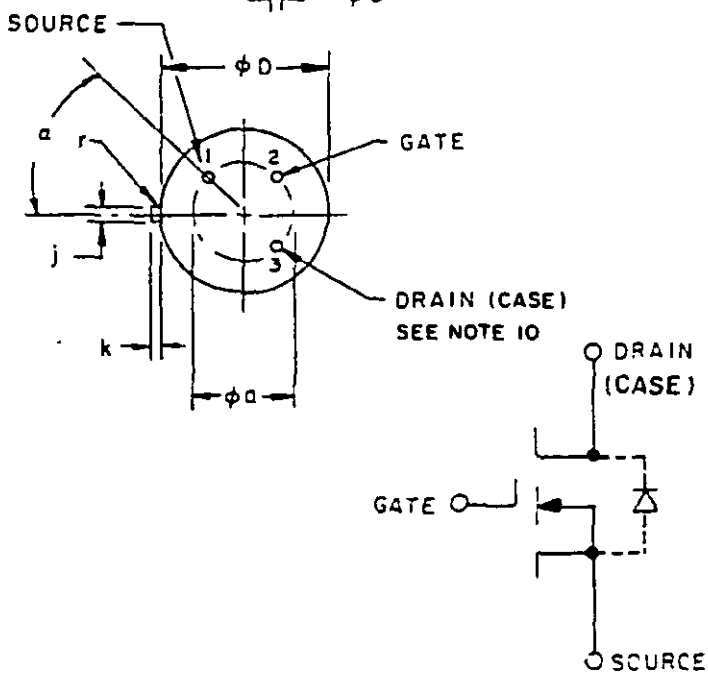
4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500. Alternate flow is allowed for qualification inspection in accordance with figure 2 of MIL-S-19500.

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SCHEMATIC CIRCUIT

Ltr	Dimensions				
	Inches		Millimeters		
	Min	Max	Min	Max	Notes
A	.240	.260	6.10	6.60	
ϕa	. 200 TP		5.08 TP		6
ϕb	.016	.021	0.41	0.53	7, 8
ϕb_1	.016	.019	0.41	0.48	7, 8
ϕD	.335	.370	8.51	9.40	
ϕD_1	.305	.335	7.75	8.51	
h	.009	.041	0.23	1.04	
j	.028	.034	0.71	1.14	2
k	.029	.045	0.74	1.14	3
L	.500	.750	12.70	19.05	7, 8
L_1	---	.050	---	1.27	7, 8
L_2	.250	---	6.35	---	7, 8
P	.100	---	2.54	---	5
Q	---	.050	---	1.27	4
r	---	.010	---	0.25	9
α	45 TP		45 TP		6

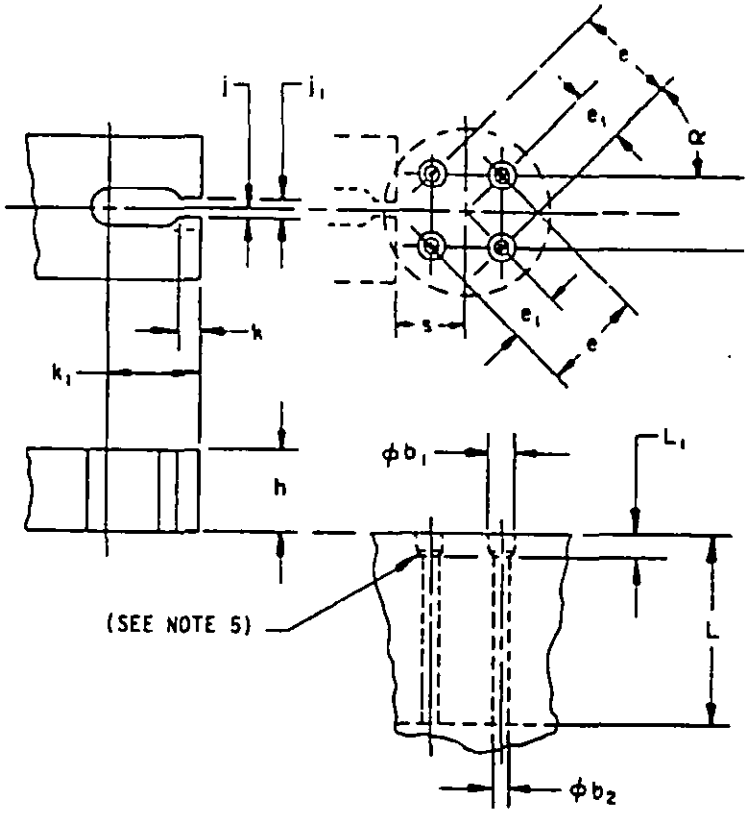
FIGURE 1. Physical dimensions for TO-205AD.

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NOTES:
1. Metric equivalents are given for general information only.
2. Beyond radius (r) maximum, j shall be held for a minimum length of .011 (0.028 mm).
3. Dimension k measured from maximum øD.
4. Outline in this zone is not controlled.
5. Dimension $øD_1$ shall not vary more than .010 (0.25 mm) in zone P.. This zone is controlled for automatic handling.
6. Leads at guage plane .054 +.001, -.000 (1.37 +0.03, -0.00 mm) below seating plane shall be within .007 (0.18 mm) radius of true position (TP) at maximum material material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by the guage and guaging procedure shown on figure 2.
7. $øb_1$ applies between L_1 and L_2. øb applies between L_2 and L minimum. Diameter is uncontrolled in L_1 and beyond L minimum.
8. All three leads.
9. Radius (r) applies to both inside corners of tab.
10. Drain is electrically connected to the case.

FIGURE 1. Physical dimensions for TO-205AD - Continued.

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Ltr	Dimensions				
	Inches		Millimeters		
	Min	Max	Min	Max	Notes
$øb_1$	.0595	.0605	1.51	1.54	
$øb_2$	.0325	.0335	0.83	0.85	
e	.1995	.2005	5.07	5.09	
e_1	.0995	.1005	2.53	2.55	
h	.150 Nominal		3.81 Nominal		
j	.0175	.0180	0.44	0.46	
j_1	.0350	.0355	0.89	0.90	
k	.009	.011	0.23	0.28	
k_1	.125 Nominal		3.18 Nominal		
L	.372	.378	9.45	9.60	
L_1	.054	.055	1.37	1.40	3
S	.182	.199	4.62	5.05	4
α	44.90°	45.10°	44.90°	45.10°	

NOTES:
1. Metric equivalents are given for general information only.
2. The location of the tab locator within the limits indicated will be determined by the tab and flange dimensions of the device being checked.
3. Gauging procedure. The device being measured shall be inserted until its seating plane is .125 ±.010 (3.18 ±0.25 mm) from the seating surface of the gauge. A force of 8 ounces ±0.5 shall then be applied parallel and symmetrical to the device's cylindrical axis and seating plane of the device shall be seated against the gauge. The use of a pin straightener prior to insertion in the gauge is permissible.
4. Gauging plane.
5. Drill angle.

FIGURE 2. Gauge for lead and tab locations.

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4.3 Screening (JANTX, JANTXV, and JANS levels only). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
9	I_{GSS1}, I_{DSS1}, Gate stress test (see 4.5.3)	Gate stress test (see 4.5.3)
10	Method 1042, test condition B	Method 1042, test condition B
11	I_{GSS1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$ ΔI_{GSS1} = ±10 nA dc or ±100% of initial value, whichever is greater. ΔI_{DSS1} = ±1 µA dc or ±100% of initial value, whichever is greater.	I_{GSS1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$
12	Method 1042, test condition A; and test condition C (see 4.3.1)	Method 1042, test condition A
13	Subgroups 2 and 3 of table I herein: ΔI_{GSS1} = ±10 nA dc or ±100% of initial value, whichever is greater. ΔI_{DSS1} = ±1 µA dc or ±100% of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20% of initial value or ±.5 ohm, whichever is greater. $\Delta V_{GS(th)1}$ = ±10% of initial value or ±0.3 V dc, whichever is greater.	Subgroup 2 of table I herein: ΔI_{GSS1} = ±10 nA dc or ±100% of initial value, whichever is greater. ΔI_{DSS1} = ±1 µA dc or ±100% of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20% of initial value or ±.5 ohm, whichever is greater. $\Delta V_{GS(th)1}$ = ±10% of initial value or ±.3 V dc, whichever is greater.

4.3.1 Power burn-in. Power burn-in conditions are as follows: T_A = +25°C; -5°C, +10°C; $V_{DS} \geq$ 10 V dc minimum; I_D - Adjusted to meet a junction temperature of 140°C, -5°C, +10°C by controlling V_{GS} voltage to obtain a specific I_D current required to apply an appropriate power for the device under stress. Power condition and T_J requirements can be established by ΔV_{SD} measurement technique described in MIL-STD-750, method 3161. For condition C, t = 240 hours.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500. Alternate flow is allowed for quality conformance inspection in accordance with figure 2 of MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. (End-point electrical measurements shall be in accordance with the applicable steps of table IV herein.)

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500, and tables IIa and IIb herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and table III herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal impedance. Thermal impedance measurements shall be performed in accordance with method 3161 of MIL-STD-750. $R_{\theta JC(max)}$ = 20°C/W.

I_M measuring current	10mA
I_H drain heating current	.41 A
t_H heating time	Steady state (see MIL-STD-750, method 3161 for definition)
V_H drain-source heating voltage	10 V
t_{MD} measurement time delay	10 to 80 µs
t_{SW} sample window time	10 µs (max)

4.5.3 Gate stress test.

V_{GS} = 30 V t = 250 µs min.

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TABLE I. Group A inspection.

Inspection	MIL-STD-750		LTPD 1/		Symbol	Limits		Unit
	Method	Conditions	JANS	JANTX, JANTXV		Min	Max	
Subgroup 1								
Visual and mechanical inspection	2071							
Subgroup 2								
Breakdown voltage, drain to source	3407	V_{GS} = 0 V; I_D = 10 µA dc; bias condition C			$V_{(BR)DSS}$			
2N6660						60	---	V dc
2N6661						90	---	V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$; I_D = 1.0 mA dc			$V_{GS(th)1}$	0.8	2.0	V dc
Gate current	3411	V_{GS} = +20 and -20 V dc bias condition C; V_{DS} = 0			I_{GSS1}	---	±100	nA dc
Drain current	3413	V_{GS} = 0 V; bias condition C; V_{DS} = 80% of rated V_{DS}			I_{DSS1}	---		
2N6660							1	µA dc
2N6661							1	µA dc
Drain to source on-state voltage	3405	Bias condition A; V_{GS} = 10 V; I_D = 1 A pulsed (see 4.5.1)			$V_{DS(on)1}$			
2N6660						---	3.0	V
2N6661						---	4.0	V
Drain to source on-state voltage	3405	Bias condition A; V_{GS} = 5 V dc; pulsed (see 4.5.1)			$V_{DS(on)2}$			
2N6660		I_D = 0.3 A dc				---	1.5	V
2N6661		I_D = 0.3 A dc				---	1.6	V
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc; 2/ bias condition A; pulsed (see 4.5.1)			$r_{DS(on)1}$			
2N6660		I_D = 1.0 A dc				---	3.0	ohm
2N6661		I_D = 1.0 A dc				---	4.0	ohm
Static drain to source on-state resistance	3421	V_{GS} = 5 V dc; 2/ bias condition A; pulsed (see 4.5.1)			$r_{DS(on)2}$			
2N6660		I_D = 0.3 A dc				---	5.0	ohm
2N6661		I_D = 0.3 A dc				---	5.3	ohm

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750 Method	MIL-STD-750 Conditions	LTPD 1/ JANS	LTPD 1/ JANTX, JANTXV	Symbol	Limits Min	Limits Max	Unit
Subgroup 2 - Continued								
Forward voltage (source drain diode)	4011	Pulsed method; V_{GS} = 0 V			V_{SD}			
2N6660		I_S = .99 A dc				0.7	1.6	V(pk)
2N6661		I_S = .86 A dc				0.7	1.4	V(pk)
Forward transconductance	3475	Pulsed (see 4.5.1)			g_{FS}			
2N6660		V_{DS} = V_{DS} = 7.5 V dc; I_{D1} = 525 mA; I_{D2} = 475 mA				170	---	mS
2N6661						170	---	mS
Subgroup 3								
High temperature operation		T_C = +125°C						
Gate reverse current	3411	Bias condition C; V_{GS} = ±20 V dc; V_{DS} = 0			I_{GSS2}	---	±100	nA dc
Drain current	3413	Bias condition C; V_{GS} = 0; V_{DS} = 80% of rated V_{DS}			I_{DSS2}	---	100	µA dc
Static drain to source on-state voltage	3405	V_{GS} = 10 V dc; Bias condition A; pulsed (see 4.5.1)			$V_{DS(on)3}$			
2N6660		I_D = 1 A dc				---	5.6	V dc
2N6661		I_D = 1 A dc				---	7.5	V dc
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc; pulsed (see 4.5.1) 2/			$r_{DS(on)3}$			
2N6660		I_D = 1.0 A dc				---	5.6	ohm
2N6661		I_D = 1.0 A dc				---	7.5	ohm
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$; I_D = 1.0 mA dc			$V_{GS(th)2}$	0.3	---	V dc
Low temperature operation		T_C = -55°C						
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$; I_D = 1.0 mA dc			$V_{GS(th)3}$	---	2.5	V dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		LTPD 1/		Symbol	Limits		Unit
	Method	Conditions	JANS	JANTX, JANTXV		Min	Max	
Subgroup 4								
Switching time test	3472	R_{GEN} = 50Ω, R_{GS} = 50Ω						
Turn-on time					$t_{(on)}$			
2N6660		$I_D \approx$ 1 A; V_{DD} = 25 V dc				---	10	ns
2N6661		$I_D \approx$ 1 A; V_{DD} = 25 V dc				---	10	ns
Turn-off time					$t_{(off)}$			
2N6660		$I_D \approx$ 1 A; V_{DD} = 25 V dc				---	10	ns
2N6661		$I_D \approx$ 1 A; V_{DD} = 25 V dc				---	10	ns
Small signal common source short-circuit input capacitance	3431	V_{DS} = 25 V dc; V_{GS} = 0 V; f = 1 MHz			C_{iss}			
2N6660						---	50	pF
2N6661						---	50	pF
Small signal common source short-circuit output capacitance		V_{DS} = 25 V dc; V_{GS} = 0 V; f = 1 MHz See figure 5			C_{oss}			
2N6660						---	40	pF
2N6661						---	40	pF
Small signal common source short-circuit reverse transfer capacitance	3433	V_{DS} = 25 V dc; V_{GS} = 0 V; f = 1 MHz			C_{rss}			
2N6660						---	10	pF
2N6661						---	10	pF

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		LTPD 1/		Symbol	Limits		Unit
	Method	Conditions	JANS	JANTX, JANTXV		Min	Max	
Subgroup 5								
Safe operating area test	3474	See figure 4						
High voltage test		V_{DS} = 80% of rated V_{DS}						
2N6660 2N6661								
Electrical measurements		See table IV; steps 1,2,3,4,5,6, and 7						
Subgroup 6								
Not applicable								
Subgroup 7								
N/A								

1/ For LTPD numbers, see MIL-S-19500.

2/ Not a measured value $r_{DS(on)} = V_{DS}$ (on)/I_D.

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TABLE IIa. Group B inspection for JANS devices.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD 1/	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 1				
Physical dimensions	2066	See figure 1		
Subgroup 2				
Solderability	2026			
Resistance to solvents	1022			
Subgroup 3				
Thermal shock (temperature cycling)	1051			
Hermetic seal	1071			
Fine leak Gross leak				
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7		
Decap, internal visual design verification	2075	See 3.3.2		
SEM	2077			
Bond strength	2037	Test condition A; all internal wires for each device shall be pulled separately.		
Die shear	2017			
Subgroup 4				
Intermittent operation life (LTPD)	1042	Test condition D. Power on time shall be 3 minutes min. 2,000 cycles. No heat sink nor forced-air cooling on the device shall be permitted during the on cycle.		
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, 7 and 12		
Subgroup 5				
Accelerated steady-state operation life	1027	120 hours; See 4.3.1 except T_J = 200°C -0°C +20°C Marking legibility requirements shall not apply		
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7		

See footnote at end of table.

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TABLE IIa. Group B inspection for JANS devices - Continued.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD 1/	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 5 - Continued				
Bond strength (Al-Au die interconnects only)	2037	Test condition A		
Subgroup 6				
N/A				

1/ For LTPD numbers see MIL-S-19500.

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TABLE IIb. Group B inspection for JANTX and JANTXV devices.

Inspection	Method	MIL-STD-750 Conditions	LTPD 1/
Subgroup 1			
Solderability	2026 App. A		
Resistance to solvents	1022		
Subgroup 2			
Thermal shock (temperature cycling)	1051		
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7.	
Subgroup 3			
Steady-state operation life (LTPD)	1027	$T_J = P_T (R_{\theta JA}) + T_A$, $P_T = V_{DS} I_D$ V_{DS} = 15 V dc $I_D \leq$ .05 A dc $R_{\theta JA}$ = 170°C/W T_A = 30°C ±5°C See figure 8	
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, 7, and 12	
Subgroup 4			
Decap, internal visual (design verification)	2075	See 3.3.2	
Bond strength	2037	Test condition A; all internal bond wires for each device shall be pulled separately.	
Subgroup 5			
Not applicable			
Subgroup 6			
High temperature life	1032	T_A = +150°C	
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7	

1/ For LTPD numbers, see MIL-S-19500.

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TABLE III. Group C inspection.

Inspection	Method	MIL-STD-750 Conditions	LTPD 1/
Subgroup 1			
Physical dimensions	2066	See figure 1	
Subgroup 2			
Thermal shock (glass strain)	1056	Test condition A	
Terminal strength (lead fatigue)	2036	Test condition E	
Hermetic seal	1071		
Fine leak Gross leak			
Moisture resistance	1021	Omit initial conditioning	
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7	
Subgroup 3			
Shock	2016		
Vibration, variable frequency	2056		
Constant acceleration	2006	30,000 G min Y1 plane only	
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7	
Subgroup 4			
Salt atmosphere (corrosion)	1041		
Subgroup 5			
Not applicable			
Subgroup 6			λ=5
Steady-state operation life	1026	$T_J = P_T (R_{\theta JA}) + T_A$ $P_T = V_{DS} I_D$; V_{DS} = 15 V dc $I_D \leq 0.05$ A dc, $R_{\theta JA}$ = 170°C/W T_A = 30°C ±5°C see figure 8	
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, 7, and 12	

See footnote at end of table.

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TABLE III. Group C inspection - Continued.

Inspection	Method	MIL-STD-750 Conditions	LTPD 1/
Subgroup 7			3
Qualification only			
Thermal shock (glass strain) destructive	1056	-55°C to 150°C 75 cycles	
Hermetic seal a. Fine leak b. Gross leak	1071		
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7	
Subgroup 8			10
Qualification only			
Steady state reverse bias	1042	Test condition A, 1,000 hrs	
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7	
Subgroup 9			10
Qualification only			
Steady state gate bias	1042	Test condition B, 1,000 hrs	
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7	

1/ LTPD numbers are to be taken from MIL-S-19500.

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TABLE IV. Groups A, B, and C electrical measurements.

Step	Inspection	MIL-STD-750 Method	MIL-STD-750 Conditions	Symbol	Limits Min	Limits Max	Unit
1.	Breakdown voltage drain to source	3407	V_{GS} = 0 V; I_D = 10.0 µA dc; bias condition C	$V_{(BR)DSS}$			
	2N6660				60	---	V dc
	2N6661				90	---	V dc
2.	Gate to source voltage (threshold)	3403	V_{DS} = V_{GS}; I_D = 1.0 mA dc	$V_{GS(th)1}$	0.6	2.0	V dc
3.	Gate current	3411	V_{GS} = +20 and -20 V dc; V_{DS} = 0 V; bias condition C	I_{GSS1}	---	100	nA dc
4.	Drain current	3413	V_{GS} = 0 V dc; bias condition C; 80% $V_{(BR)DSS}$	I_{DSS1}			
	2N6660		V_{DS} = 48 V dc		---	1	µA dc
	2N6661		V_{DS} = 72 V dc		---	1	µA dc
5.	Static drain to source on-state resistance	3421	V_{GS} = 10 V dc; bias condition A, pulsed (see 4.5.1) 1/	$r_{DS(on)1}$			
	2N6660		I_D = 1.0 A dc		---	3.0	ohm
	2N6661		I_D = 1.0 A dc		---	4.0	ohm
6.	Drain to source on-state voltage	3405	V_{GS} = 5 V dc; bias condition A, pulsed (see 4.5.1)	$V_{DS(on)}$			
	2N6660		I_D = 0.3 A dc		---	1.5	V dc
	2N6661		I_D = 0.3 A dc		---	1.6	V dc
7.	Forward voltage (source drain diode)	4011	Pulsed method, see 4.5.1 V_{GS} = 0 V	V_{SD}			
	2N6660		I_S = .99 A dc		0.7	1.6	V(pk)
	2N6661		I_S = .86 A dc		0.7	1.4	V(pk)
8.	Gate current	3411	Bias condition C; V_{GS} = ±15 V dc; V_{DS} = 0 V dc	ΔI_{GSS}	±10 nA dc or 100% of initial value, whichever is greater		
9.	Drain cutoff current	3413	Bias condition C; V_{DS} = 90 V dc for 2N6661 V_{GS} = 0 V dc V_{DS} = 60 V dc for 2N6660	ΔI_{DSS}	±1 µA dc or 100% of initial value, whichever is greater		
10.	Small signal, drain to source on-state resistance	3423	V_{GS} = 10 V dc; I_D = 1.0 A dc; 1/ f = kHz; pulsed, see 4.5.1	$\Delta r_{DS(on)}$	±0.5 ohm or ±20% of initial value, whichever is greater		
11.	Gate to source threshold voltage	3403	V_{DS} = V_{GS}; I_D = 1.0 mA dc	$\Delta V_{GS(th)}$	±10% of initial value or ±0.3 V dc, whichever is greater		
12.	Thermal resistance	3161	See 4.5.2	$R_{\theta JC}$	---	20	°C/W

1/ Not a measured value $r_{DS(on)} = V_{DS(on)}/I_D$

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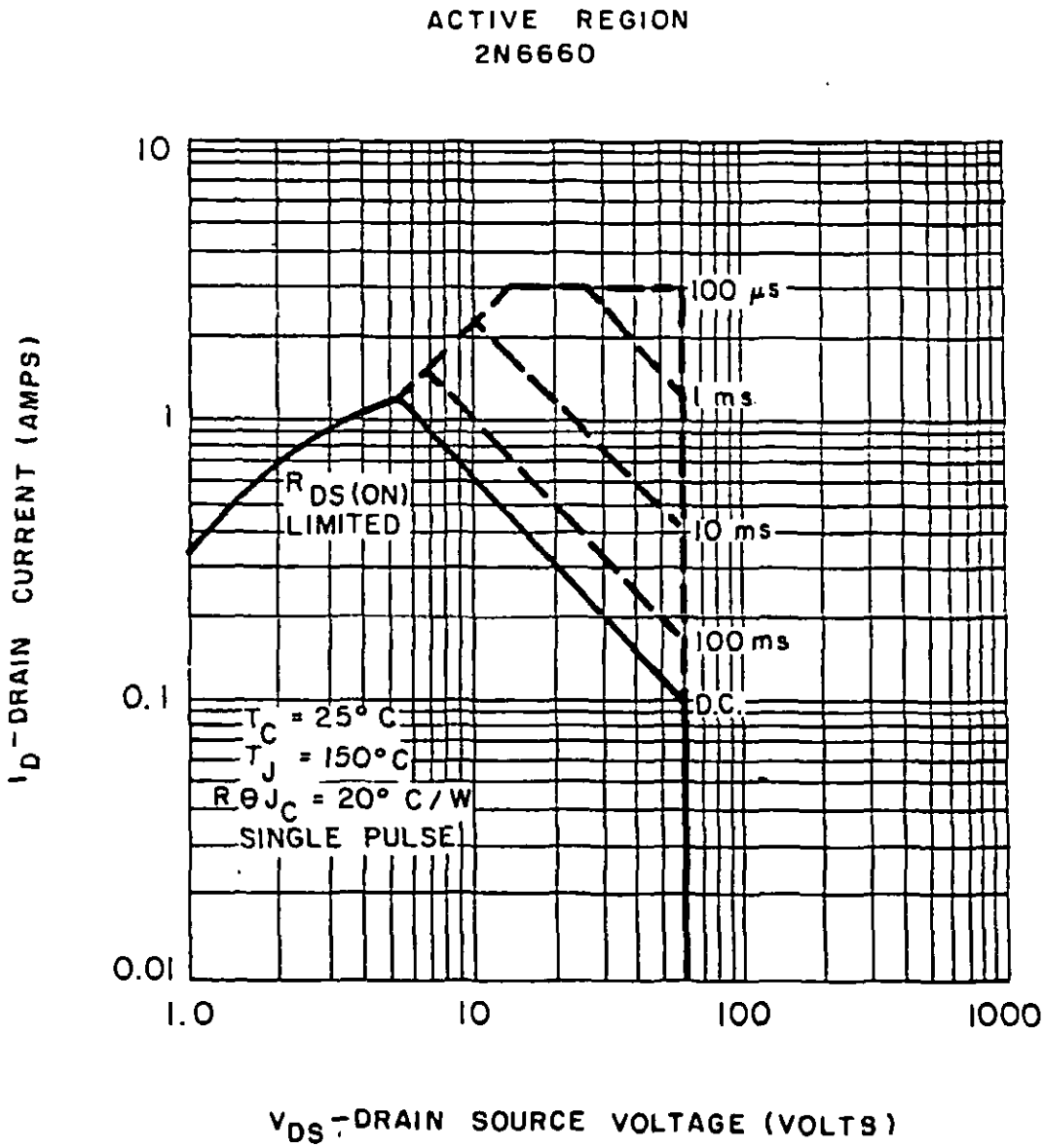


FIGURE 4. Maximum safe operating area.

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ACTIVE REGION

2N6661

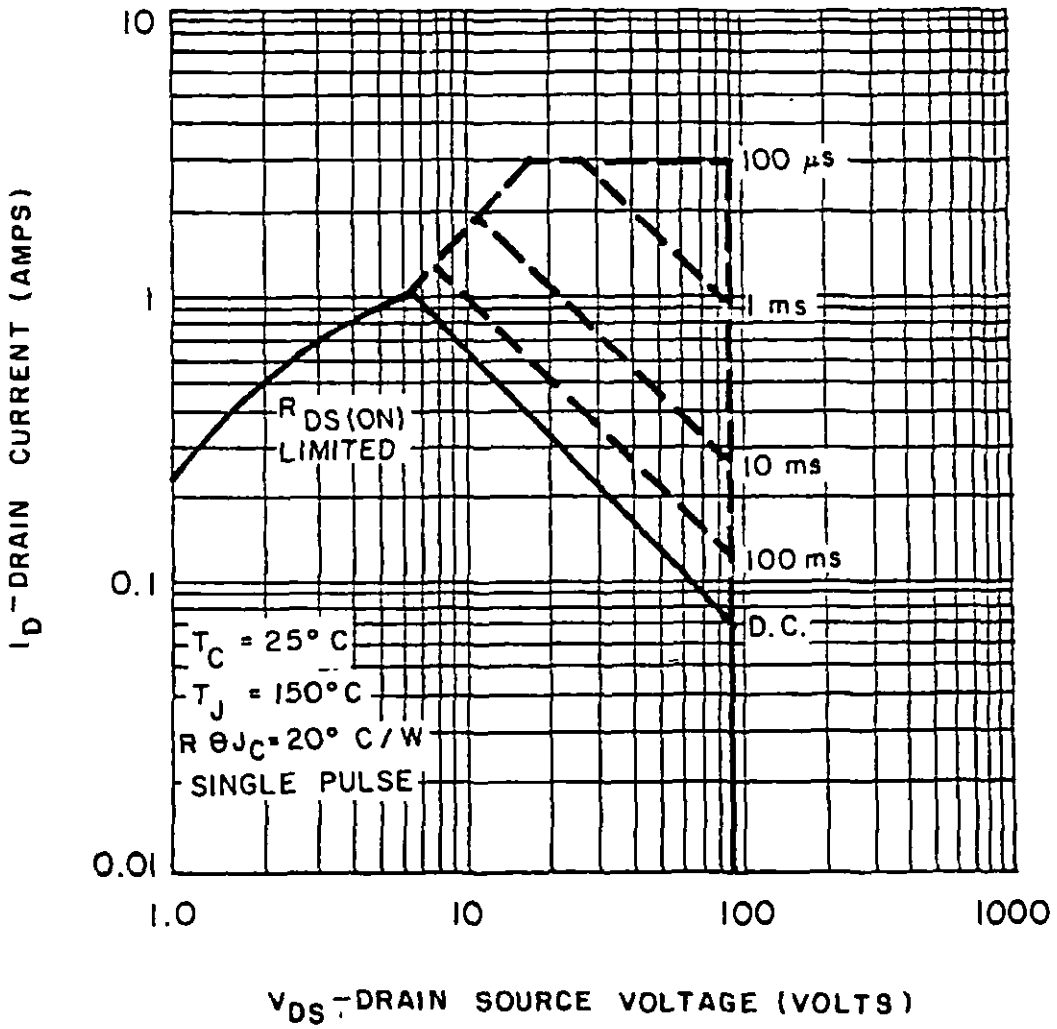
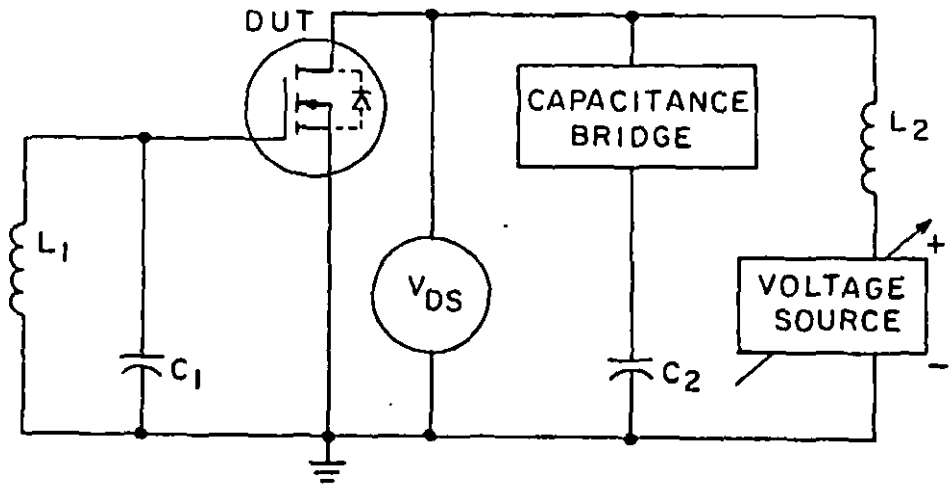


FIGURE 4. Maximum safe operating area - Continued.

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NOTE: The capacitors C_1 and C_2 shall present short circuits at the test frequency. L_1 and L_2 shall present a high ac impedance at the test frequency for isolation. The bridge shall have low dc resistance between its output terminals and shall be capable of carrying the test current without affecting the desired accuracy of measurement.

FIGURE 14. Test circuit for small signal, common source, short-circuit, output capacitance.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. However, the following handling practices are recommended (see 3.5).

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

g. Care should be exercised during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, R $\leq$100 K, whenever bias voltage is to be applied drain to source.

6.3 Ordering data. Acquisition documents may specify the material and finish (see 3.3.1).

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Navy - TD
Air Force - 11, 70, 80
NASA - EB13

User activity:
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
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	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

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